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INSTITUTE OF ENGINEERING
PULCHOWK CAMPUS

A REPORT ON
NETWORK DESIGN
(Mini Project)

Submitted To:
DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING
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Network Design for XYZ Software Company

This project outlines a computer network design for XYZ Software Company. It will use technologies like VLAN, OSPF, DHCP, and more to improve how different parts of the company's network communicate and share information.

Project Objectives

1. Create a network with efficient routing and connections using OSPF.
2. Set up separate areas for each department using VLANs.
3. Install special switches and access points in different areas.
4. Arrange automatic IP addresses using DHCP services.

Network Topology

The software company has been allocated the IP address range of 10.0.0.0/22 to establish a computer network infrastructure. The company is composed of various departments, each characterized by specific device counts:

1. Software Development: 300 devices

2. Sales and Marketing: 200 devices

3. Quality Assurance: 175 devices

4. Design and User Experience: 150 devices

5. Customer Support: 130 devices

6. Administration and HR: 80 devices

7. Research and Innovation: 75 devices

8. Recreation: 50 devices

9. Project Management: 50 devices

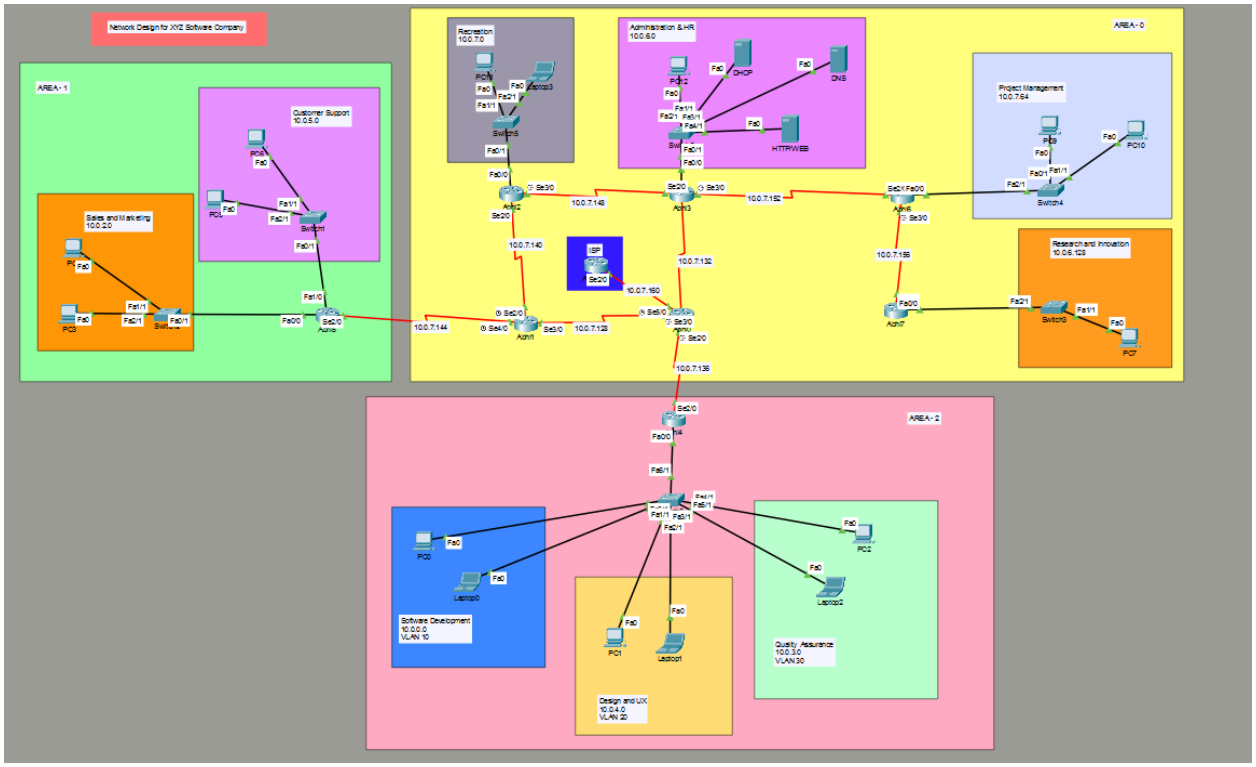

Fig : Network Topology for XYZ Software Company

Network Devices

The topology consists of the following devices:

- **Routers** used for routing between different networks and OSPF areas.
- **Switches** used to connect multiple devices within each department.
- **PCs and Laptops** representing users in different departments.
- **Servers** providing DNS, DHCP, and web services.
- **ISP Router** providing Internet connectivity.

IP Address Pool

The company has been assigned the following IP address block:

10.0.0.0/22

Total available addresses: **1024**

Address range:

10.0.0.0 – 10.0.3.255

This address block is divided into multiple subnets using **VLSM (Variable Length Subnet Mask)** to accommodate different department sizes.

Subnet Table

Departments	**Network ID**	**Subnet Mask**	**Host Range**	**Broadcast Address**
Software Development	10.0.0.0	255.255.254.0 / 23	10.0.0.1 - 10.0.1.254	10.0.1.255
Sales and Marketing	10.0.2.0	255.255.255.0 / 24	10.0.2.1 - 10.0.2.254	10.0.2.255
Quality Assurance	10.0.3.0	255.255.255.0 /24	10.0.3.1 - 10.0.3.254	10.0.3.255
Design and UX	10.0.4.0	255.255.255.0 /24	10.0.4.1 - 10.0.4.254	10.0.4.255
Customer Support	10.0.5.0	255.255.255.0 /24	10.0.5.1 - 10.0.6.254	10.0.5.255
Administration and HR	10.0.6.0	255.255.255.128 /25	10.0.6.1 - 10.0.6.126	10.0.6.127

Research and Innovation	10.0.6.128	255.255.255.128 /25	10.0.6.129 - 10.0.6.254	10.0.6.255
Recreation	10.0.7.0	255.255.255.192 /26	10.0.7.1 - 10.0.7.62	10.0.7.63
Project Management	10.0.7.64	255.255.255.192 /26	10.0.7.65 - 10.0.7.126	10.0.7.127

Network Description

Networks : There are 9 routers having different sizes.

Routers: There are 9 routers in total . Abhi0-Abhi7 and Abhi_ISP

Routing : OSPF is used for internal routing . Also, all internet traffic has been forwarded to upstream service provider. There are three OSPF areas.

OSPF Area	**Depatments Included**
Area 1	Sales and Marketing , Customer Support
Area 2	Software Development, Design and UX , Quality Assurance
Area 0	Rest of the network

Servers : There are DHCP server , DNS server and web server in the Administration and HR Dept.

VLAN

Three separate VLANs are extended across the network using switches to provide distinct communication environment for different departments.

Department	VLAN
Software Development	10
Design and ux	20
QA	30

Routing Strategy

Open Shortest Path First (OSPF) protocol will be used. This will allow for dynamic routing, adapting to changes in the network.

Managing IP Addresses

To make things simpler , routers are configured to act as a DHCP server. This dynamically assigns IP addresses, subnet masks , gateway addresses and other network configuration parameters to devices that connect to the network.

Results and Observations

The network design project yielded the following results:

- By implementing DHCP, IP addresses were assigned dynamically to devices within the network.
- During the project, the "ping" and "tracert" commands were utilized to observe the connectivity and trace the routes of data packets.
- Each internal network could be reached from any computer or laptop within the setup.
- The routing mechanisms performed as anticipated.

Conclusion

The intended network design for the software development company seeks to create a secure and resilient communication framework. By intelligently deploying VLANs, OSPF routing, automatic IP allocation, and dedicated access points, this network is focused on enhancing collaboration and resource distribution across different departments and areas within the workspace. This project offers valuable insights into real-world applications of networking concepts.